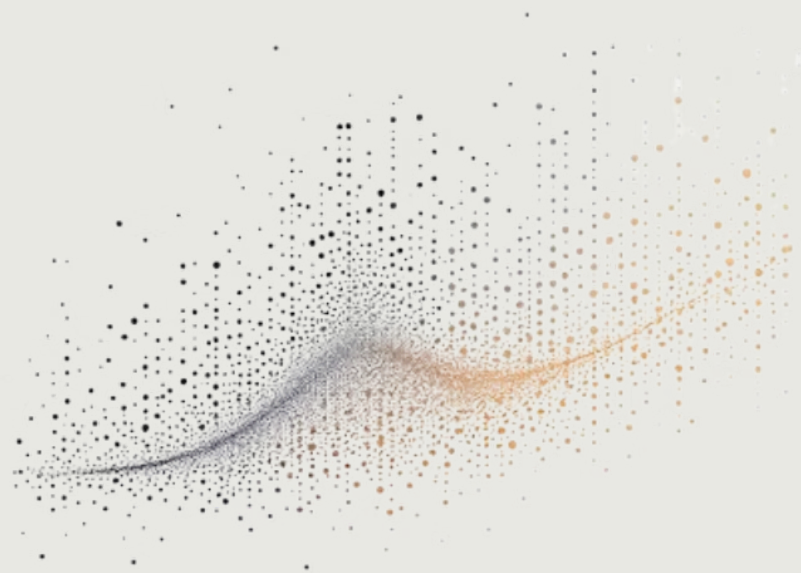


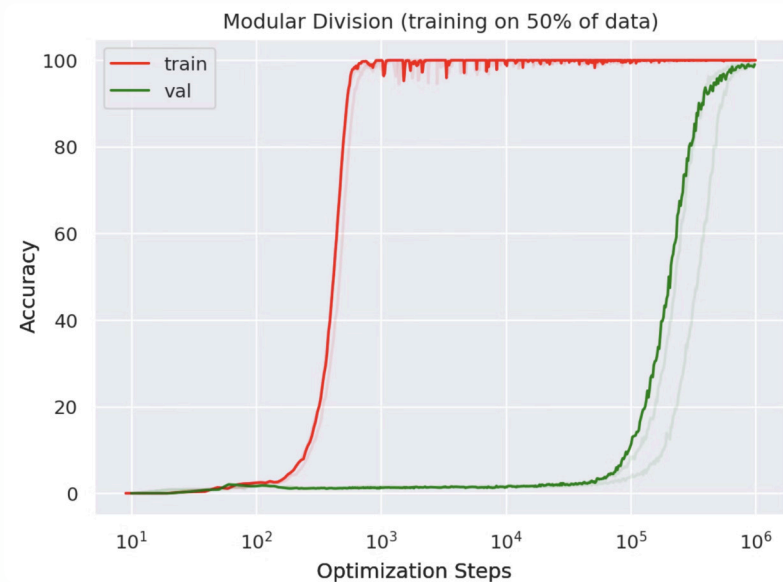
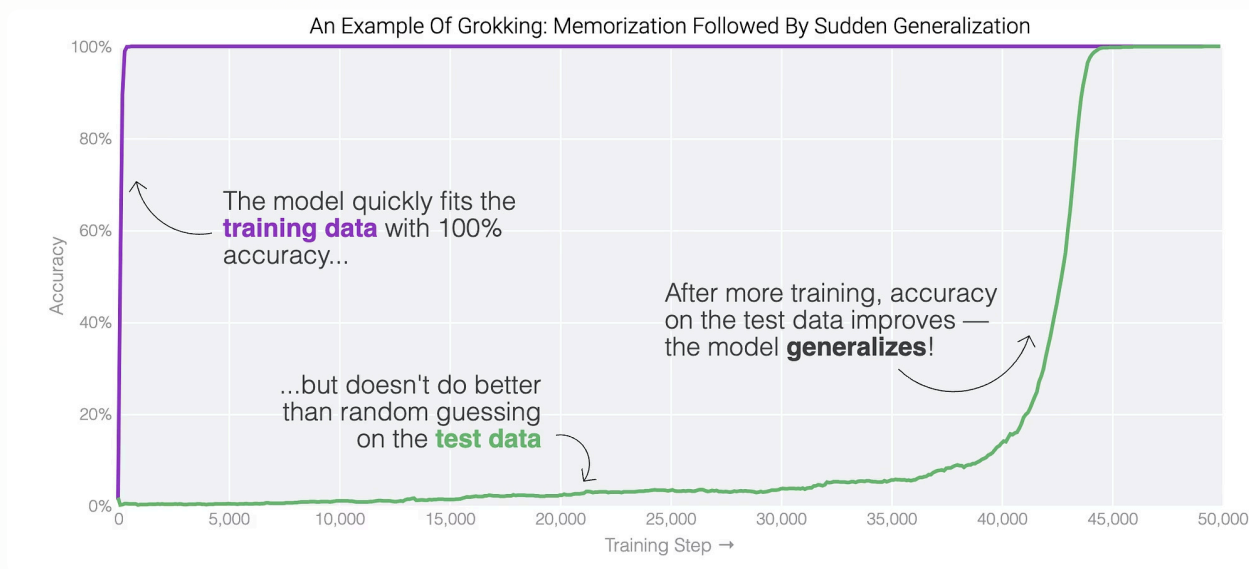


Grokking Across Models, Benchmarks, and LLMs

Austin Tsai, Freddie Liang, Peter Wang, and Ryan Nene

Test Accuracy Changes after Plateauing

Even after test accuracy plateaus, generalization suddenly occurs after many iterations!



Research Question



When does Grokking occur?

Across synthetic datasets and real-world data



Real ML contexts?

How does it apply in practical machine learning scenarios?



Why This Matters

01

Learning dynamics

Understanding how models learn over time

03

Dataset structure

How data organization affects learning patterns

02

Architecture caveats

Model-specific behaviors and limitations

04

Real-world applicability

Practical implications for ML applications

Previous Work

Grokking: Generalization Beyond Overfitting on Small Algorithmic Datasets - Power et al.

- Study that coined the term Grokking and showed that it occurred.
- Eventually models learn the rules and not just memorizes solutions

Grokking Explained: A Statistical Phenomenon - Carvalho et al.

- When there is a different distribution between test and train data, more time is needed to learn the underlying pattern

Predicting Grokking Long Before it Happens - Notsawo et al.

- Early detection of grokking through training loss in first few epochs
- Reduces the time needed to study Grokking by predicting whether or not Grokking will occur

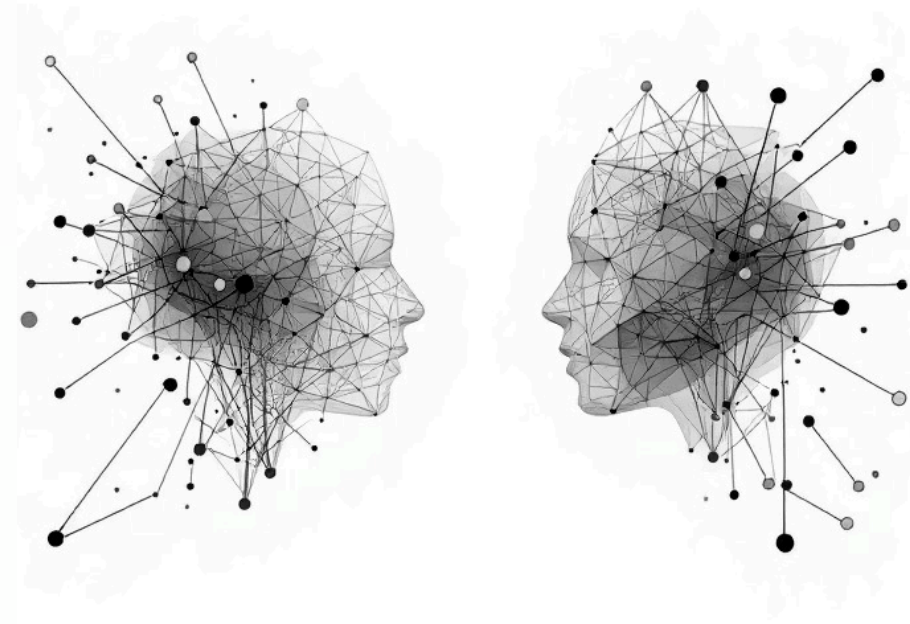
Our Approach

Previous Limitations

- Mostly small transformer architectures
- Few select tasks
- Theoretical without quantitative benchmarks
- Synthesized datasets only

Our Contribution

Test across different architectures and optimization algorithms to see which are more prone to grokking.



Algorithms/Models to Test

MLPs

Multilayer perceptrons

Small Transformers

Compact transformer models

Foundation Transformer
Models

Qwen4B, Gemma, etc.

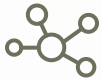
We plan to compare across:



Dataset structure



Model size (parameters)



Architectures



Regularization



No. training steps



Tasks

Datasets



Synthetic structured dataset

as seen in previous works (generated)



Language Q&A dataset → OpenAI MMMLU

from huggingface (open source)



Instruction dataset → Huggingface H4 Ultrachat 200k

from huggingface (open source)

Experiments

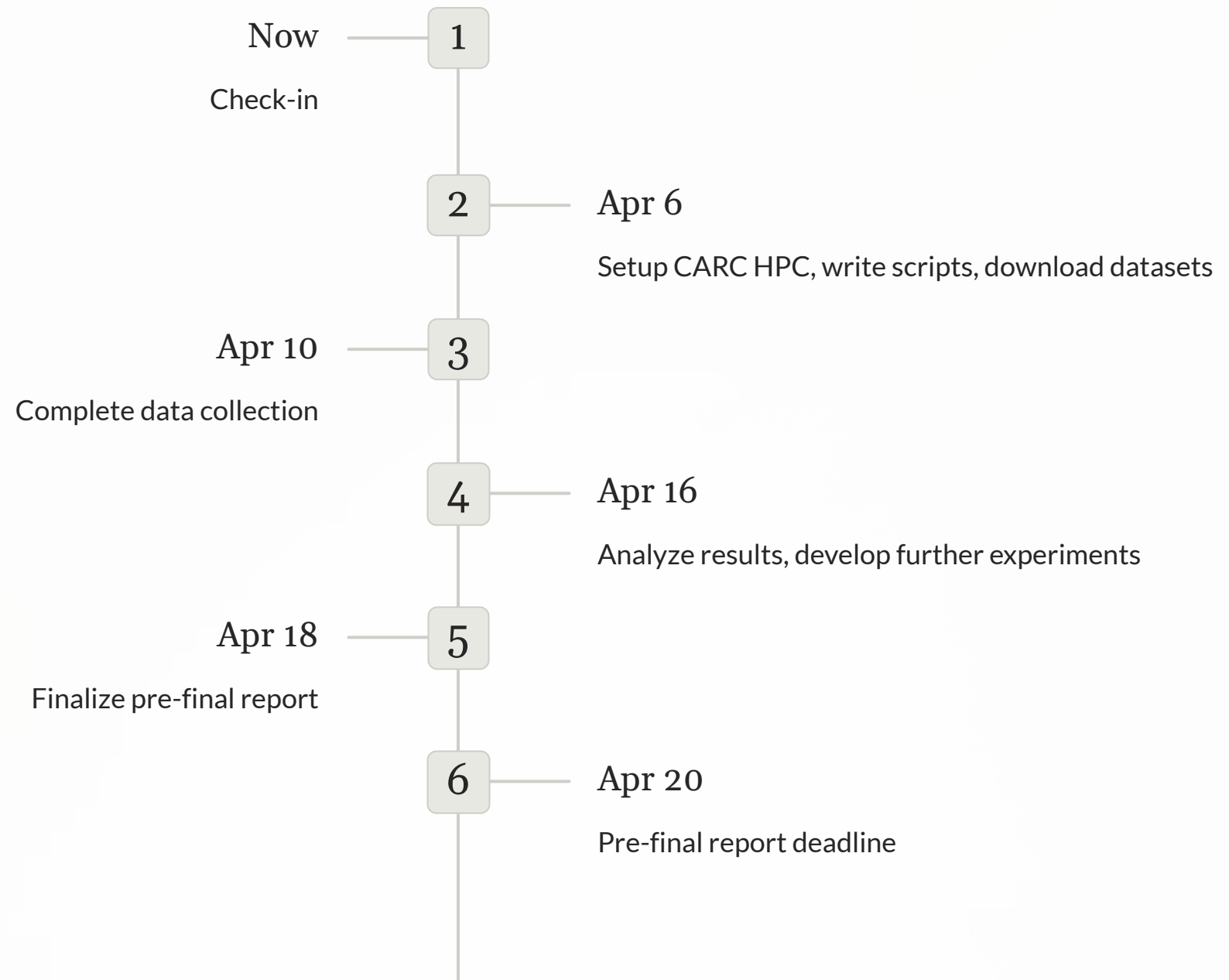
Core metric: generalization gap over training runs

- Steps at which grokking occurs
- Rate of generalization once it begins

We will try to vary variables one at a time to isolate what triggers grokking

Variable	Values
Architecture	MLP, small transformer, foundation LLM (stretch)
Dataset	Synthetic, HuggingFace Q&A, instruction dataset
Model size	Small / medium (vary # parameters)
Regularization	Weight decay off / low / high
Optimizer	Adam / AdamW (standard)
Training steps	Extended horizon (grokking requires long runs)

Timeline & Next Steps





Q&A and Discussion

We welcome your questions, insights, and feedback on our research proposal!